

SAILAKSHMI B T

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PROFESSIONAL SUMMARY

Robotics and Automation undergraduate (CGPA 8.99) at Madras Institute of Technology with hands-on experience in embedded systems, robotics, and AI-driven applications. Skilled in Arduino-based system design, sensor interfacing, and rapid prototyping. Interests span Computer Vision, industrial automation, embedded intelligence, and intelligent systems.

EDUCATION

B.E. - Robotics and Automation | Madras Institute of Technology | 2024 - 2028

- CGPA: 8.99 / 10.0 (current)
- Higher Secondary Certificate (2024): 94.16% | Secondary School Certificate (2022): 95.2%
- JEE (Main) 2024: 81.82 percentile (CRL: 269,607 | OBC-NCL: 94,238)

INTERNSHIPS & TRAINING

Computer Vision & AI Intern | PixelBotix | June 2026

- Completed hands-on training in image processing, object detection, and AI-based visual recognition; developed and deployed small-scale CV/AI projects.

Industrial Robotics & Automation 1-day Training | Birla Carbon & KUKA | 28th Feb 2026

- Trained on industrial robotic arm programming, automation workflows, and safety practices; exposed to Industry 4.0 concepts and KUKA robot operation.

PROJECTS

Student Life Concierge AI Agent | Google ADK, Gemini API, FastAPI, Python, Pytest, Antigravity CLI | Kaggle & Google 5-Day AI Agents Intensive

- Built a multi-agent FastAPI backend with prompt-injection defenses and a Pytest mock-service suite to automate student scheduling using Google ADK, Gemini API and Antigravity CLI.

Masters Application Tracker & Resume Analyzer | React, Vite, PDF.js, LocalStorage API, Antigravity CLI

- Built a privacy-first, client-side app with a PDF.js resume analysis engine for keyword extraction, profile evaluation and a Masters application tracker for Undergraduate Engineering students.

Orion - Local AI Assistant | Python, Llama 3.2, Ollama, FastAPI, WebSockets, Antigravity CLI

- Developed an offline assistant using Llama 3.2 via Ollama with an async FastAPI backend and WebSocket streaming. Currently expanding tool-calling capabilities for local system diagnostics.

Single-Axis Gimbal Stabilization System | Arduino Nano, MPU-6050, SG90 Servo | Control Systems

- Built a closed-loop PID gimbal stabilizer using an MPU-6050 IMU and embedded C on Arduino Nano for real-time angular correction.

Grip Stability & Stress Analyzer | Arduino Uno, HX711, Load Cell | Biomedical device

- Built a load-cell measurement system with HX711 ADC to capture grip force and tremor patterns for stress-level estimation.

Adaptive Self-Mutating Authentication System | 8051 (AT89S52), Embedded C

- Designed an 8051-based digital lock that regenerates a non-deterministic password each login, resisting replay attacks and shoulder-surfing.

Mechanical Rotary Device | SolidWorks | Design Patent Pending

- Designed a dual-mode Geneva wheel/dog clutch rotary mechanism in SolidWorks; filed for design patent.

HACKATHONS & COMPETITIONS

Smart India Hackathon 2025 | Delhi, Finalist | 2025

- Collaborated on reverse-engineering multi-architecture firmware binaries to identify cryptographic primitives and protocols via AI/ML.

H2TS Hackathon 2025 | 2nd Place | 2025

- Collaborated with a cross-functional team to build and deploy an AI-driven SOS platform for rapid emergency response.

TECHNICAL SKILLS

Embedded Systems: Arduino Uno R3, Arduino Nano, 8051 (AT89S52), Sensor Interfacing, Hardware Timers, I2C/SPI Communication

Control Systems and Robotics: PID Control, Closed-Loop Feedback, Motion Control, Inertial Sensing, Stabilization Mechanisms, ROS2

Programming: Python (Intermediate), Embedded C (Fundamentals), MATLAB/Simulink (Intermediate)

AI & Agentic Systems: Google Agent Development Kit (ADK), Gemini API, Ollama, Llama 3.2, FastAPI, Antigravity CLI, Prompt Engineering

Computer Vision: Image Processing, Object Detection, AI-based Visual Recognition (OpenCV-level coursework)

Design & EDA: SolidWorks, KiCAD, Canva

Cybersecurity / RE: Binwalk, Ghidra, Firmware Binary Analysis, Ubuntu (working knowledge of tools and workflow)

CERTIFICATIONS

- Automation in Manufacturing - IIT Guwahati, NPTEL (Elite, 67%)
- AI Capabilities and Limitations - Anthropic
- Foundations of Prompt Engineering - AWS Training & Certifications
- Python for Absolute Beginners - Udemy
- HTML, CSS, & JavaScript Certification Course for Beginners - Udemy

LEADERSHIP & ACTIVITIES

- **Placement Representative** | *2026-Present* | Liaison between students and placement coordinators.
- **Joint Treasurer, MIT Quill** | *2025-26* | Managed treasury for the college literary club.
- **Office Bearer & Content Writer, SAHA** | *2025-26* | Organized academic events, Science and Humanities Association.
- **Member, ARME** | *2024-Present* | Association of Robotics and Mechatronics Engineers.
- **YRC Volunteer, MIT** | *2024-Present* | Organized Youth Fest events for visually challenged individuals.
- **Assistant School Pupil Leader** | *2019-20* | Directed administrative support for student body operations and school event coordination.
- **Master of Ceremonies** | *2017-2020* | Hosted multiple school events to develop public speaking and audience engagement.

LANGUAGES

English (Proficient) | Tamil (Proficient) | Telugu (Conversational) | Japanese (Elementary) | Hindi (Elementary)